

# Hanyeol Ryu

Research Assistant

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## RESEARCH INTERESTS

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- Low-power wireless communications
- Channel modeling & estimation
- AI/ML techniques for intelligent wireless communications
- Wireless sensing
- Hardware-Software co-designed IoT applications

## EDUCATION

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### Pusan National University

B.S. in Electronics Engineering

Feb 2026

- PI: Prof. Sangkil Kim

## RESEARCH EXPERIENCE

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### Pusan National University

Undergraduate Researcher (PI: Dr. Sangkil Kim)

Jan 2024 – Feb 2026

- Develop and evaluate energy-efficient backscatter communication systems through hardware–software co-design, enhancing channel reliability and enabling scalable multimedia-oriented IoT applications.

Research Assistant (PI: Dr. Sangkil Kim)

Mar 2026 – Present

- Investigate deep learning-driven channel estimation and signal recovery techniques for ultra-low-power backscatter and integrated sensing–communication systems under severe hardware constraints.

## PUBLICATIONS

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### Published Articles (†: First author, \*: Corresponding author)

- **J3:** N. Ha, **H. Ryu**, H. Jeong, H. Kim, G. Kim, A. Georgiadis, M. M. Tentzeris, and S. Kim\*, “Integrated Sensing and Backscatter Communication (ISABC): Toward Self-Sustainable Autonomous IoT Networks,” *IEEE Microwave Magazine*, Under Review.
- **J2:** **H. Ryu**<sup>†</sup>, and S. Kim\*, “Ultra-low-power Monostatic Backscatter Platform with Phase-Aware Channel Estimation and System-Level Validation,” *IEEE Internet of Things Journal*, Accepted, May 2026.
- **J1:** J. Choi<sup>†</sup>, **H. Ryu**<sup>†</sup>, and S. Kim\*, “Improved Backscattering Communication Range Using Reflection Amplifiers,” *J. Korean Inst. Electromagn. Eng. Sci*, vol. 36, no. 2, pp. 266–273, Feb. 2025.
- **C1:** **H. Ryu**<sup>†</sup>, and S. Kim\*, “High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System,” in *Proc. IEEE Antenna Propag. Int. Symp.*, 2025, pp. 2877–2879.

## PATENTS

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- Korean Patent: “Residual Phase-Aware Channel Estimation Method for Monostatic Backscatter Communication Systems”, Apr 2026.

## CONFERENCE PRESENTATIONS

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**Title:** Ultra-low-power Backscatter Platform with Phase-Aware Channel Estimation (Oral)

- The Korean Institute of Electromagnetic Engineering and Science (KIEES) Feb 2026

**Title:** High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System (Oral)

- IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (AP-S/URSI) Jul 2025

**Title:** Improved Backscattering Communication Range Using Reflection Amplifiers (Poster)

- The Korean Institute of Electromagnetic Engineering and Science (KIEES) Feb 2025

## HONORS & AWARDS

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- 2<sup>nd</sup> Place: Hackathon (Theme: Improving Accessibility for People with Disabilities and the Elderly), **Pusan National University**, Aug 2023.
- 2<sup>nd</sup> Place: Capstone Design Project, **Pusan National University**, Nov 2025.

## SERVICE & EXTRACURRICULAR ACTIVITIES

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### University

**Pusan National University**

*Treasurer, Student Council*

Dec 2021 – Dec 2022

*Student Ambassador, Pusan National University*

Apr 2021 – Feb 2023

*Volunteer, Overseas Volunteering Program (3D Printer & Arduino) — Thailand*

Dec 2022 – Feb 2023

### Military Service

**Republic of Korea Marine Corps**

*Squad Leader, ROK Marine Corps Force Reconnaissance*

Feb 2017 – Jul 2018

2<sup>nd</sup> Place, **ROK Marine Corps Force Reconnaissance Selection and Qualification Training**

Jun 2017 – Aug 2017

## SKILLS & TECHNIQUES

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- **Programming:** MATLAB, Python, C++
- **AI/ML:** PyTorch, TensorFlow/Keras
- **RF/EM Simulation:** Ansys HFSS, Advanced Design System (ADS)
- **SDR Platform:** GNU Radio
- **Hardware Design:** OrCAD PCB Editor, OrCAD Capture
- **3D Modeling:** Rhino 7